

Prototype a Grounded Assistant

Project: Verified Study Buddy

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Block to Python Bridge

Without opening the old lesson, explain one key idea from Block to Python Bridge. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Tools and APIs

Without opening the old lesson, explain one key idea from Tools and APIs. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Retrieval Augmented Generation

Without opening the old lesson, explain one key idea from Retrieval Augmented Generation. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the worked case for Capstone grounded assistant. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Capstone grounded assistant. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply a new case independently

Request room J; approval and register both J capacity 0. Draft is well-formed with J/0 and correct sources. Trace stages and distinguish zero from missing.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

Build a paper assistant from notes A='Roots take up water' and B='Leaves use light'. Answer 'Which part takes up water?' and 'How tall is this plant?' with references or fallback.

6. Solve the given case

Approved notes say plants need light and roots take up water. Ask how roots help.

Specify the minimum grounded path.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

A study assistant has a supplied passage about roots. A user asks a question about planets. What should the prototype do within a passage-only boundary?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 24 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Connect user interface, source store, retrieval, prompt, schema validation, and evidence display.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

Build a paper assistant from notes A='Roots take up water' and B='Leaves use light'. Answer 'Which part takes up water?' and 'How tall is this plant?' with references or fallback.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

My reasoning and evidence

13. W2 Trace the worked case

Request room B; retrieved approval names B but register names C. Draft gives capacity 10 from C. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

14. W3 Audit the limit

A report omits this qualification: A correct happy path does not establish fallback, access control or side-effect safety. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

15. W4 Apply a new case independently

Request room J; approval and register both J capacity 0. Draft is well-formed with J/0 and correct sources. Trace stages and distinguish zero from missing.

My reasoning and evidence

16. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Prototype a Grounded Assistant

Build a minimal grounded study answer with checks between retrieval and display.

Source PL2: seedlings need water. Source PL3: this activity checks soil daily. Question: what does this activity check daily?

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Prototype a Grounded Assistant

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Retrieve evidence

PL3 directly supports the daily soil-check claim; PL2 supplies a different fact.



2. Match question

Choose the activity's soil check, not a general watering recommendation.



3. Validate draft

Display "This activity checks soil daily [PL3]"; reject an invented twice-daily watering schedule.

CHANGE ONE THING

PL3 is absent while PL2 remains.

Explain the changed result using the matching lesson.

Can PL2 alone support "check the soil daily"? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Verified Study Buddy, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

19. Project portfolio milestone

Project: Verified Study Buddy

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

For your Verified Study Buddy, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Build a minimal grounded study answer with checks between retrieval and display.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Display “This activity checks soil daily [PL3]”; reject an invented twice-daily watering schedule.

1. Retrieve evidence
2. Match question
3. Validate draft

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Validate draft Display “This activity checks soil daily [PL3]”; reject an invented twice-daily watering schedule. Read the picture in order. Display “This activity checks soil daily [PL3]”; reject an invented twice-daily watering schedule. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed-case practice: PL3 is absent while PL2 remains. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. Explain that the daily check is not specified in the available source.
3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Explain that the daily check is not specified in the available source. The remaining general water fact does not establish this activity's schedule.

Repair a missing building block

Review Block to Python Bridge (ai-3/23). Goal: Translate a short paper calculation into a Python script. Short practice: Trace the changed list and identify where the result differs. Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Tools and APIs (ai-3/21). Goal: Explain the mechanism and terminology of tools and api contracts. Short practice: Does a valid C7 ID permit the extra lending action? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Retrieval Augmented Generation (ai-3/18). Goal: Explain the mechanism and terminology of building a grounded answer. Short practice: Which part of the source supports a two-minute melting claim? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

